MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Exam Type EXAMINATION Month YYYY
Semester IV
DSE – MG4DSEZGYA01 – AN UNDERSTANDING OF RIVERINE ECOLOGY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Choose the Correct Option

Answer all questions

Each question carries **1** marks

1. A continuous record of discharge plotted against time is called_____ [U] / [1]
   a) Stream flow
   b) Flow pulse
   c) Hydrograph
   d) Flow regime

2. An area drained by a river and its tributaries is_____ [U] / [1]
   a) Catchment area
   b) River basin
   c) Streams
   d) Tributary

3. Which of the following is not a key factor in Anthropocene river classification [K] / [2]
   a) Sediment load
   b) Water temperature
   c) Water cycle
   d) Riparian Zone

4. DPSIR Framework denotes [K] / [2]
   a) Drivers, Provision, State, Innovate, Response
   b) Division, Protect, System, Impact, Reuse
   c) Drivers, Pressure, System State, Impact, Response
   d) Division, Productive, System, Impact, Response

5. Repair, restore and protect are done as a result of [U] / [2]
   a) Drivers
   b) Impact
   c) Pressure

d) Response

6. Select the ideal range of TDS for drinking water [U] / [2]
a) 200-300
b) 10-20
c) 50-150
d) 201-250

7. The provisioning services of rivers does not include [U] / [3]
a) Consumptive use
b) Non- consumptive use
c) Abiotic materials
d) Erosion control

8. The River Research Centre in Kerala is situated in [K] / [4]
a) Pathanamthitta
b) Thrissur
c) Kottayam
d) Palakkad

9. Which of the following is not a river tourism village in Kerala. [U] / [5]
a) Kumbalangi
b) Malarikkal
c) Puzhakkal
d) Pykara

10. Which of the following is not a suitable type for cage culture in rivers. [A] / [3]
a) Fixed
b) Aerial
c) Floating
d) Submersible & Submerged

[1x10 = 10]

Part B

Short Answer Questions

Answer any **14** questions

Each question carries **2** marks

11. Write a short account on Indicators of Hydrologic Alteration. [U] / [1]

12. Differentiate tributaries and distributaries. [U] / [1]

13. Differentiate stream cascades and reaches. [U] / [1]

14. Summarize the key characteristics of a stream glide. [U] / [1]

15. What are invertivores? [U] / [1]

16.	What are the key impacts of land use on streamflow?	[U] / [2]
17.	Mention any four driving forces that leads to pressure on environment	[U] / [2]
18.	Name four impacts caused by humans on biodiversity	[U] / [2]
19.	Give two examples for regulating services of river	[U] / [3]
20.	Mention safe limits of river water for drinking.	[An] / [2]
21.	What are Rain Gauges?	[U] / [2]
22.	Distinguish BOD and COD.	[U] / [2]
23.	Give a short account on any one river restoration in Kerala	[U] / [3]
24.	Mention any two pollution abatement works taken up under NRCP	[U] / [3]
25.	How to avail Janakeeya Malsya Krishi(KMK)?	[I] / [3]
26.	Name any two River Conservation Organisations in Kerala and their functions	[U] / [4]

[2x14 = 28]

Part C

Short Essay Questions

Answer any **8** questions

Each question carries **4** marks

27.	Classify mineral substrates in the riverbed based on particle size.	[U] / [1]
28.	What is longitudinal profile of a river? Give its key characteristics	[U] / [1]
29.	Discuss the key roles of macroinvertebrates in riverine ecosystem.	[U] / [1]
30.	Write an account on River Continuum Concept(RCC)	[U] / [1]
31.	Elaborate on the man made threats to the river.	[A] / [2]
32.	Explain the impacts of altered bio-geo chemical cycles on biodiversity	[U] / [2]

33. Explain briefly the methods to assess microbes in rivers [U] / [2]

34. Describe pen culture methods followed in rivers [U] / [3]

35. Discuss the major research areas related to riverine ecosystem [U] / [3]

36. Describe how cage culture is carried on in rivers [I] / [5]

[4x8 = 32]

QPCode

Reg.No : .....

Name : .....